

AI-based Air Force ATO Production Support System

RunTime Error!

Capt J. Park, Systems Division, Command and Communications Department, ROK AFOC
1st Lt Y. Kim, Systems Division, Command and Communications Department, ROK AFOC (Currently @ KAI)
SSgt S. Kim, Systems Division, Command and Communications Department, ROK AFOC (Currently @ Sungkyunkwan Univ.)
SrA J. Song, Combat Operations Team, Combat Operations Division, ROK AFOC (Currently @ Carnegie Mellon Univ.)

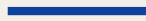
I will now begin Team RUNTIME ERROR's presentation on an AI-based ATO production support system.

>> Table of Contents

1	Overview
2	Idea Implementation
3	Refinement Plan
4	Q&A

We will proceed in the following order: Overview, Idea Implementation, our Refinement Plan, and Q&A.

1



Concept and Current State of ATO

Before we start, *please be aware that the data mentioned hereafter may differ from actual facts.*

First, let's look at the concept of an ATO and its current state.

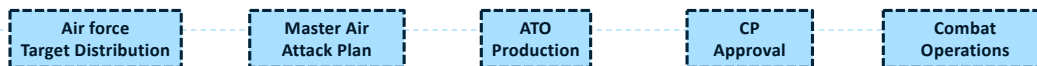
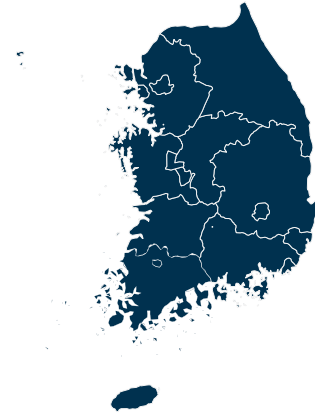
ATO (Air Tasking Order)

항공임무명령서

- ✓ **Definition:** A document detailing all air sorties within the airspace, including targets, missions, and operational plans
- ✓ **Purpose:** A tool for efficiently controlling air force power during joint operations under the Combined Forces Commander

"An Air Tasking Order (ATO) is a powerful tool that aids the Combined Air Component Commander to efficiently and effectively plan, organize and direct air operations through centralized planning."

- United States 7th Air Force



An ATO, or Air Tasking Order, is a document that specifies the targets, missions, and operational plans for all aircraft sorties within our airspace, serving as a means to effectively control the Air Force's air power.

Air Force aerial operations are carried out through the series of steps shown below. First, when a target is assigned to the Air Force—in other words, when a mission is issued—we match the appropriate aircraft to execute that mission and draft an MAAP. Then, we produce the ATO in the form of a timetable for each base, and once command approval is given, we create the actual sorties through the combat operations process.

1. Overview (1-1/3) - The Concept of ATO

Airbase	Type	Mission	Armory	Mission_Location	Time_On_Target	Target_Location	Mission_Altitude	Dept_Time	RTB_Time
K-2	F-15K	CAP	AGM-65	Daejeon	15:35:00	36.3684° ,127.3668°	12-20	15:30:00	15:42:20

** This is a fictional example and does not reflect actual events.*

Air force
Target Distribution

Master Air
Attack Plan

ATO
Production

CP
Approval

Combat
Operations

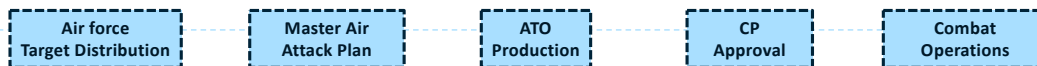
Here is an example of an ATO. Can you guess what it means? (3s pause)

1. Overview (1-2/3) - Real-World Examples (Simulated)

Airbase	Type	Mission	Armory	Mission_Location	Time_On_Target	Target_Location	Mission_Altitude	Dept_Time	RTB_Time
K-2	F-15K	CAP	AGM-65	Daejeon	15:35:00	36.3684° ,127.3668°	12-20	15:30:00	15:42:20

Deploy an **F-15K fighter** equipped with **AGM-65** from **K-2 Daegu Air Base** at **15:30**
and conduct **patrol missions** near **Daejeon Creative Economy Innovation Center** between **12,000-20,000** ft at 15:35.
(Place where the presentation was taken)

* This is a fictional example and does not reflect actual events.



This single line indicates that,
from Daegu Air Base,
an F-15K fighter jet equipped with the AGM-65 weapon
is to take off at 15:30
to perform a patrol mission
at an altitude of 12,000 to 20,000 feet
near the Daejeon Center for Creative Economy and Innovation. (Where this presentation is taking place)

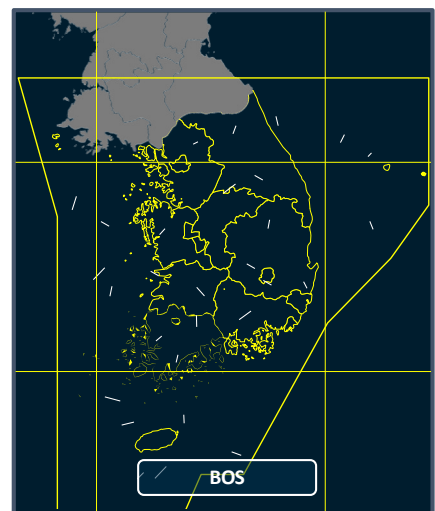
That single line is **just part of one ATO**.
Even that alone is quite complex, **so imagine how much information needs to be consolidated to create hundreds or even thousands of such ATOs.**

1. Overview (1-3/3) - The Concept of ATO

A	B	C	D	E	F	G	H	I
1	Airbase Type	Mission	Army	Location	Strike Time	Target Location	Altitude	RDV Time
2	C-130	Interd	AI	JDAU	20R	03:10	125 84 37 537	10-18
3	C-119	F-16	AI	AQAP-BS	12D	03:58	125 788 37 583	8-28
4	C-130	F-16	BSAD	AQAP-BS	10D	13:15	125 744 37 474	10-18
5	C-119	F-16	BSAD	AQAP-BS	20R	02:41	125 425 37 581	11-23
6	C-130	Interd	AI	AQAP-BS	12D	01:47	125 868 37 585	12-09
7	C-130	Interd	AI	AQAP-BS	12D	07:54	125 584 37 587	11-23
8	C-130	Interd	AI	AQAP-BS	12D	11:10	125 402 37 544	11-19
9	C-130	Interd	AI	AQAP-BS	12D	04:27	125 037 37 54	8-28
10	C-130	Interd	AI	AQAP-BS	12D	01:30	125 891 37 546	10-25
11	C-130	Interd	AI	AQAP-BS	12D	11:17	125 853 37 515	11-24
12	C-130	Interd	AI	AQAP-BS	12D	04:50	125 84 37 5	12-18
13	C-130	Interd	AI	AQAP-BS	12D	02:30	125 884 37 526	8-21
14	C-130	Interd	AI	AQAP-BS	12D	04:44	125 482 37 52	8-18
15	C-130	Interd	AI	AQAP-BS	12D	11:12	125 149 37 534	10-18
16	C-130	Interd	AI	AQAP-BS	12D	13:22	125 542 37 5	13-28
17	C-130	Interd	AI	AQAP-BS	12D	10:34	125 862 37 487	10-20
18	C-130	Interd	AI	AQAP-BS	12D	02:44	125 85 37 575	8-16
19	C-130	Interd	AI	AQAP-BS	12D	04:14	125 488 37 544	8-28
20	C-130	Interd	AI	AQAP-BS	12D	13:09	125 258 37 52	8-17
21	C-130	Interd	AI	AQAP-BS	12D	18:47	125 374 37 511	12-28
22	C-130	Interd	AI	AQAP-BS	12D	03:38	125 131 37 5	7-29
23	C-130	Interd	AI	AQAP-BS	12D	18:30	125 138 37 548	8-24
24	C-130	Interd	AI	AQAP-BS	12D	11:58	125 009 37 129	10-20
25	C-130	Interd	AI	AQAP-BS	12D	12:35	125 545 37 581	12-25
26	C-130	Interd	AI	AQAP-BS	12D	04:50	125 864 37 552	8-28
27	C-130	Interd	AI	AQAP-BS	12D	18:27	125 639 37 583	11-25
28	C-130	Interd	AI	AQAP-BS	12D	18:18	125 04 37 5	11-25
29	C-130	Interd	AI	AQAP-BS	12D	12:14	125 639 37 512	12-28
30	C-130	Interd	AI	AQAP-BS	12D	15:33	125 684 37 581	8-21
31	C-130	Interd	AI	AQAP-BS	12D	10:25	125 663 37 585	11-19
32	C-130	Interd	AI	AQAP-BS	12D	08:20	125 42 37 587	8-16
33	C-130	Interd	AI	AQAP-BS	12D	04:43	125 037 37 528	8-16
34	C-130	Interd	AI	AQAP-BS	12D	18:21	125 402 37 515	14-27
35	C-130	Interd	AI	AQAP-BS	12D	04:58	125 333 37 528	8-16
36	C-130	Interd	AI	AQAP-BS	12D	11:37	125 168 37 581	10-28
37	C-130	Interd	AI	AQAP-BS	12D	12:14	125 82 37 5	11-24
38	C-130	Interd	AI	AQAP-BS	12D	04:47	125 639 37 483	8-24
39	C-130	Interd	AI	AQAP-BS	12D	08:31	125 82 37 589	10-24
40	C-130	Interd	AI	AQAP-BS	12D	03:26	125 932 37 519	8-25
41	C-130	Interd	AI	AQAP-BS	12D	08:08	125 241 37 589	8-24
42	C-130	Interd	AI	AQAP-BS	12D	08:28	125 521 37 515	11-17
43	C-130	Interd	AI	AQAP-BS	12D	03:38	125 54 37 528	11-28
44	C-130	Interd	AI	AQAP-BS	12D	07:05	125 844 37 543	8-28
45	C-130	Interd	AI	AQAP-BS	12D	18:47	125 283 37 415	8-20
46	C-130	Interd	AI	AQAP-BS	12D	17:05	125 337 37 585	10-28
47	C-130	Interd	AI	AQAP-BS	12D	16:08	125 102 37 488	10-27
48	C-130	Interd	AI	AQAP-BS	12D	15:15	125 713 37 527	8-18
49	C-130	Interd	AI	AQAP-BS	12D	12:19	125 257 37 521	8-22
50	C-130	Interd	AI	AQAP-BS	12D	03:03	125 525 37 527	10-28
51	C-130	Interd	AI	AQAP-BS	12D	08:25	125 844 37 582	8-27
52	C-130	Interd	AI	AQAP-BS	12D	11:27	125 884 37 583	8-19
53	C-130	Interd	AI	AQAP-BS	12D	03:38	125 54 37 528	11-28
54	C-130	Interd	AI	AQAP-BS	12D	07:05	125 844 37 543	8-28
55	C-130	Interd	AI	AQAP-BS	12D	18:47	125 283 37 415	8-20
56	C-130	Interd	AI	AQAP-BS	12D	17:05	125 337 37 585	10-28
57	C-130	Interd	AI	AQAP-BS	12D	16:08	125 102 37 488	10-27
58	C-130	Interd	AI	AQAP-BS	12D	15:15	125 713 37 527	8-18
59	C-130	Interd	AI	AQAP-BS	12D	12:19	125 257 37 521	8-22
60	C-130	Interd	AI	AQAP-BS	12D	03:03	125 525 37 527	10-28

EXCEL List

Report Slides



* This is a random example and does not reflect actual events.

* BOS, Battlefield Operating System

Air force
Target Distribution

Master Air
Attack Plan

ATO
Production

CP
Approval

Combat
Operations

In reality, the personnel who produce these ATOs must consider and incorporate a significant amount of information every day, including...

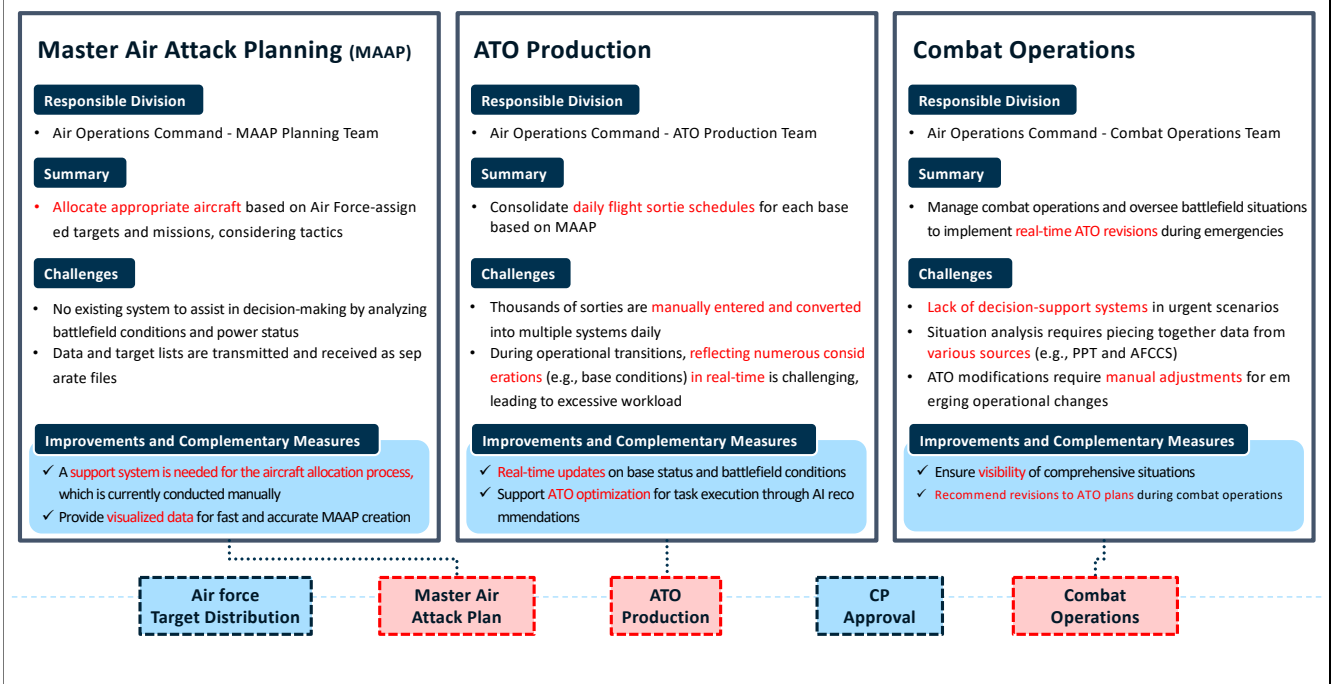
(on the left) target lists containing hundreds or thousands of these lines,

(in the center) dozens or hundreds of base status reports, and

(on the right) battlefield situation maps.

It's an *extremely labor-intensive process* that demands a *tremendous amount of time and effort* every day.

1. Overview (2/3) - Limitations of Current ATO Production

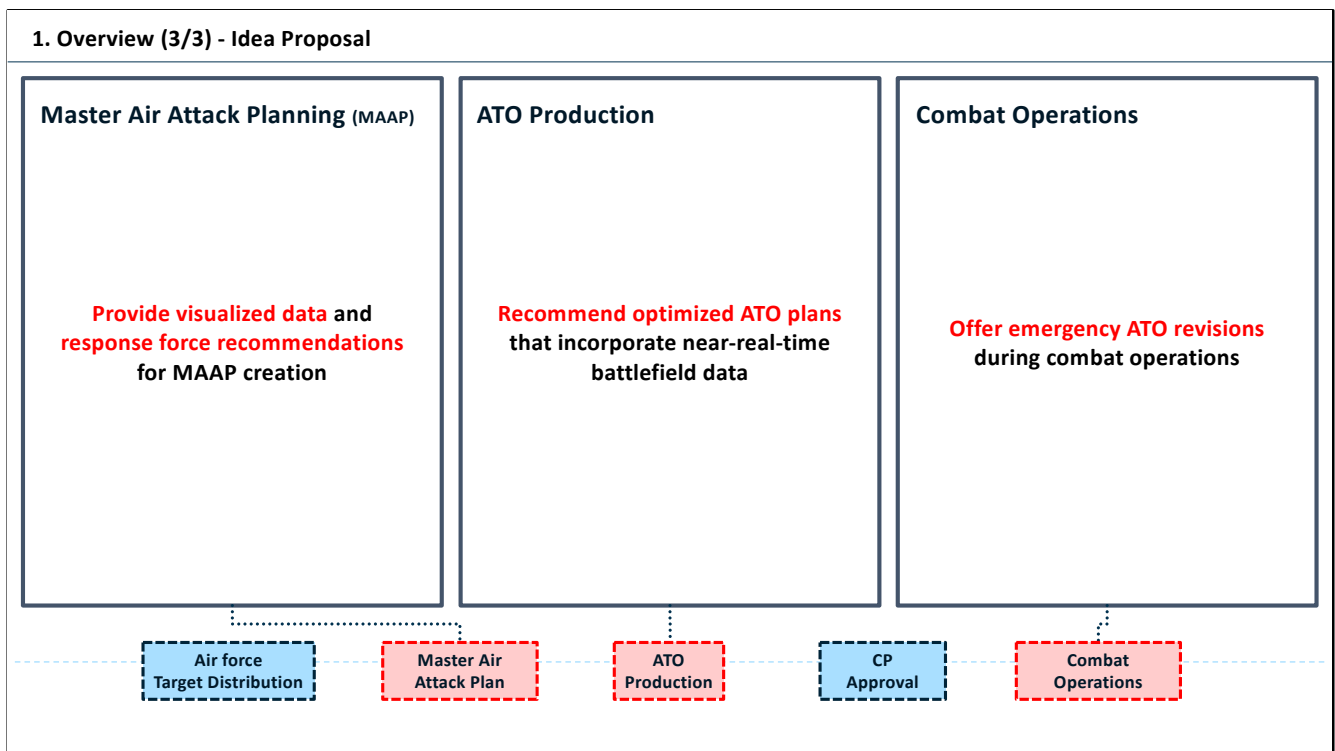


As mentioned before, there are several challenges in the current ATO production process:

(left) During the MAAP creation stage -where the Air Force's assigned targets and missions must be matched to the appropriate aircraft type with tactical considerations in mind- there is no support system that integrates and visualizes battlefield conditions, force status, and other information in a way that can be quickly and easily understood at a glance.

(center) In the ATO production stage, which involves creating daily flight schedules for each base based on the drafted MAAP, **hundreds or even thousands of sorties must be manually entered** into the ATO distribution system each day. Moreover, a multitude of factors, such as base conditions, are not updated in real time, so any change that arises directly translates into an increased workload for the combat operations team during actual missions.

(right) Finally, in the combat operations stage -where overall operations are managed and the ATO may need to be modified- there is no system to support rapid decision-making in time-critical situations. As a result, personnel must gather data on the battlefield environment piecemeal from various sources (e.g., PPT, AFCCS, phone calls) in order to make decisions.



To address these challenges, we considered what functionalities are needed during each stage and, after careful deliberation, reached the following conclusions:

(left) MAAP Creation Stage: The ability to provide visualized data and recommend suitable forces

(center) ATO Production Stage: The ability to recommend an optimized ATO that reflects real-time battlefield conditions

(right) Combat Operations Stage: The ability to recommend an alternative ATO revision plan in the event of an emergency

1. Overview (3/3) - Idea Proposal

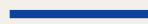
AI-based Air Force ATO Production Support System



Therefore, we propose an **AI-based ATO production support system** that provides all of these functionalities.

By reflecting our constantly changing available forces -affected by unpredictable variables- and incorporating the commander's operational objectives, this system would **produce ATOs in real time and support command decision-making** through the aid of artificial intelligence.

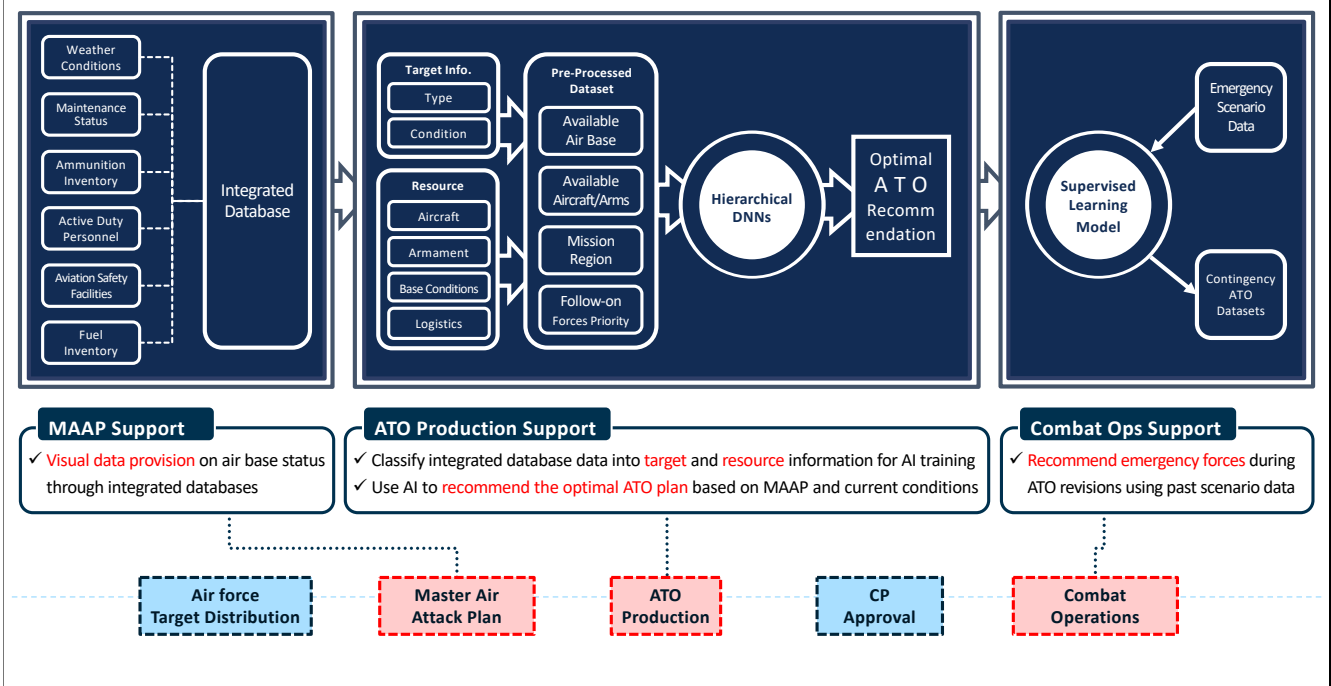
2



Idea Implementation

Next, let's discuss how we plan to implement this idea.

2. Implementation of the Idea (1/5) - Detailed System Design



Briefly put, here's the concept:

(left) Through an **integrated database**, we either input or link the operational status of each wing. This lets us calculate available forces, and then visualize those forces along with target information to support MAAP creation.

(center) Using **reinforcement learning AI that incorporates operational considerations** such as target data and available forces, we recommend an optimized, realistic ATO proposal to assist with ATO production.

(right) If the ATO needs to be revised during the operation, we use **supervised learning AI trained on existing real-world data** to recommend emergency response forces, thus supporting combat operations.

When these three functions are integrated, they form a single system that supports the entire series of Air Force aerial operations.

2. Implementation of the Idea (2/5) - Base Status Integration

MAAP Support

- Consolidate base statuses and display conditions using multiple colors
- Input real-time data on base, aircraft, and logistics supply statuses from subordinate squadrons to update the integrated database in real time

→ Utilize data in conjunction with the ATO production support system

Base Asset Status (Aircraft)

기타상황
대구기지
항공기

F-15K 001

임무 수행중

배정임무: AI - 병행 지대공 미사일
무장현황: 2 X AIM-9, 5 X JDAM
현재연료: 75%

KF10 007

임무 수행중

배정임무: XINT - 공중 대기
무장현황: 2 X AIM-9, 4 X JDAM
현재연료: 80%

F-35A 001

임무 수행중

배정임무: XINT - 핵무기 GPS교란
무장현황: 2 X AIM-9, 5 X JDAM
현재연료: 85%

KF-21 보라매 001

정비중

배정임무: 없음
무장현황: 없음
현재연료: 0%

FA-50 023

임무 대기

배정임무: GCAS - 비상대기
무장현황: 2 X AIM-9, 2 X LGB, 2 X JDAM
현재연료: 90%

KF-21 보라매 006

이륙 대기

배정임무: GCAS
무장현황: 2 X AIM-9, 4 X CLUSTER
현재연료: 95%

Base Status

기타상황
대구기지
항공기
KF-21 보라매 006

Details

KF-21 보라매 006

상태: 정상

사양

배정임무: GCAS
무장현황: 2 X AIM-9, 4 X CLUSTER
현재연료: 95%

사양

내부 연료 탱크: 8,000L
외부 연료 탱크: 10,000L
무장 탑재량: 8,000kg
최고속도: 12,000km/h
최고고도: 15,000m
최대 추력: 18,000kgf
최대 기동성: 2,000m/s²
무장 종류: 2 X AIM-9, 4 X CLUSTER

Activities

Activity	Time
목적 상공, 7면 적탐지 미션 (담당: 홍시 이준재, 935-6653)	2023. 6. 19 10:41
필요정보 이상탐지 (담당: 홍시 이준재, 935-6653)	2023. 6. 19 10:27
2 X AIM-9, 4 X CLUSTER 포집 완료 (담당: 홍시 이준재, 935-6653)	2023. 6. 19 10:28
장공중 12,000m 주입 완료 (담당: 홍시 이준재, 935-6653)	2023. 6. 19 10:30
2 X AIM-9, 4 X CLUSTER 상공 순찰 (담당: 홍시 이준재, 935-6653)	2023. 6. 19 10:42
KF-21 보라매 006 임무 종료 (담당: 홍시 이준재, 935-6653)	2023. 6. 19 10:45
목적 대기 (담당: 홍시 이준재, 935-6653)	2023. 6. 19 10:41
일정: 소정 임무종료: 9:30-9:52	

상태 입력하기

입력

I will now explain our system by each function.

First is the feature that provides an integrated view of base status for each wing.

The main screen, as shown on the left, is presented in a table format that **displays the status of essential elements needed to generate sorties for each wing**, including aircraft, runways, and navigational aids. Each status is derived from the condition of the underlying assets that make up these essential elements, with its operational impact assessed and displayed. The overall status of each wing is then represented by consolidating all the statuses of these essential elements.

For example, if you select the aircraft section of the Daegu Air Base, the top-right portion of the screen shows the current status of the aircraft there. If you then select the sixth KF-21 Boramae, the bottom-right portion displays basic information such as its assigned missions, armaments, current condition, and specifications, while also allowing you to input and review its mission and action history.

This base status information can be entered **manually** by the respective personnel or **automatically** fed into the system if there is an existing management system in place -such as the 'Smart Air Base project'- that can be integrated. It is then linked to the ATO production support systems, **forming the foundation for producing an ATO** that reflects real-time friendly force status.

2. Implementation of the Idea (2-1/5) - MAAP Support

The screenshot displays the MAAP Support interface. On the left is a dark sidebar with a menu including 'AI 기반 ATO 생산 보조체계', '메뉴', '기지상황', 'MAAP 작성', 'ATO 생산', and '전투문열'. Below the menu is a '금원의 업무' section with a message about 147 targets and a 'MAAP 작성하기' button. The main area features a map of Korea with numbered target locations (1-20). A date selector at the top shows 'MAAP 작성 2023-09-20'. On the right, a '표적 입력' section has buttons for '표적 선택' and '선택된 표적 없음'. Below it is a 'MAAP 작성' table with columns for selection, target code, base, aircraft type, weapon, and mission time. The table lists four targets: GS94, HG29, WS19, and NP76, each with associated aircraft and mission times. A 'Program Features' box at the bottom right lists: North Korea Region (target locations/info), South Korea Region (base status/aircraft readiness), User Interaction (mouse clicks/drag-and-drop), and Warnings (pop-up alert messages).

선택	표적코드	기지	기종	무장	타격시간	
☐	GS94	충주	KF-16	JDAM	오전 05:29	삭제
☐	HG29	강릉	KF-21	LGB	오전 06:15	삭제
☐	WS19	대구	F-15K	GBU-39	오전 07:41	삭제
☐	NP76	청주	F-35	JDAM	오전 08:49	삭제

Program Features

- North Korea Region:** Displays target locations/information
- South Korea Region:** Displays base status, aircraft readiness
- User Interaction:** MAAP creation through mouse clicks and drag-and-drop functionality
- Warnings:** Pop-up alert messages for mission constraints

This is the MAAP support function.

When you upload targets, the system displays their locations and information on a map.

Targets that affect air operations -such as surface-to-air missiles- are marked with red circles indicating detection and engagement ranges.

It also uses different colors to represent each base's status, and when you select a base, it shows the relevant status information and available forces.

Let's look at the prototype demonstration.

After uploading an Excel file containing the targets assigned to the Air Force, the system categorizes them by type and displays their locations on the map.

When you select a specific target, you can choose recommended strike assets suggested by the AI, which will show you the corresponding forces available at different bases.

Once you select the appropriate forces, they are added to the MAAP table on the right, and you can set the mission time accordingly.

2. Implementation of the Idea (3/5) - ATO Production

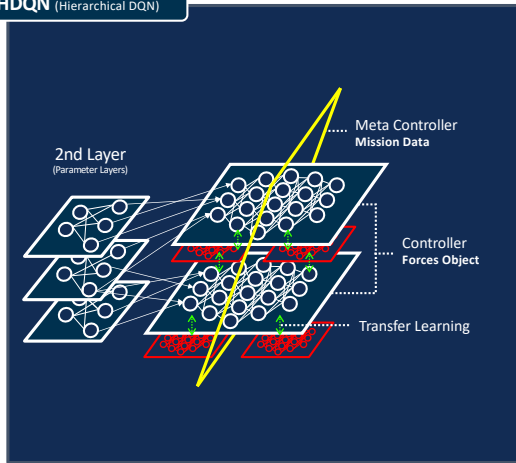


This is the ATO production support function.

Building on the results of the MAAP creation process, when generating the ATO file for distribution to subordinate wings, it utilizes information such as target types and status -as well as data linked to base conditions like aircraft, armaments, and weather. Through a hierarchical deep neural network–based reinforcement learning AI model, it then provides an optimized ATO recommendation.

2. Implementation of the Idea (4/5) - AI Explanation

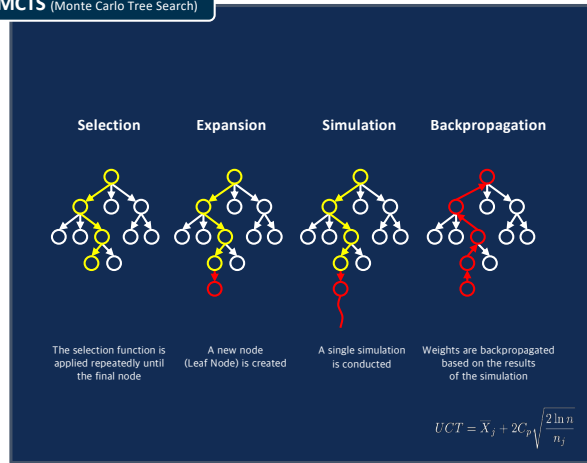
HDQN (Hierarchical DQN)



Explanation

- Reinforcement learning algorithm that solves final goals by achieving sub-goals
- Suitable for complex, multi-variable scenarios like real-world simulations
- Effective for ATO production in sparse reward environments

MCTS (Monte Carlo Tree Search)



Explanation

- Search algorithm that **selects high-probability success moves** through random exploration
- **UCT**: Adjusts the balance between selecting the best-known move and exploring new moves
→ **Prioritizes high-success strategies and can handle unpredictable situations**

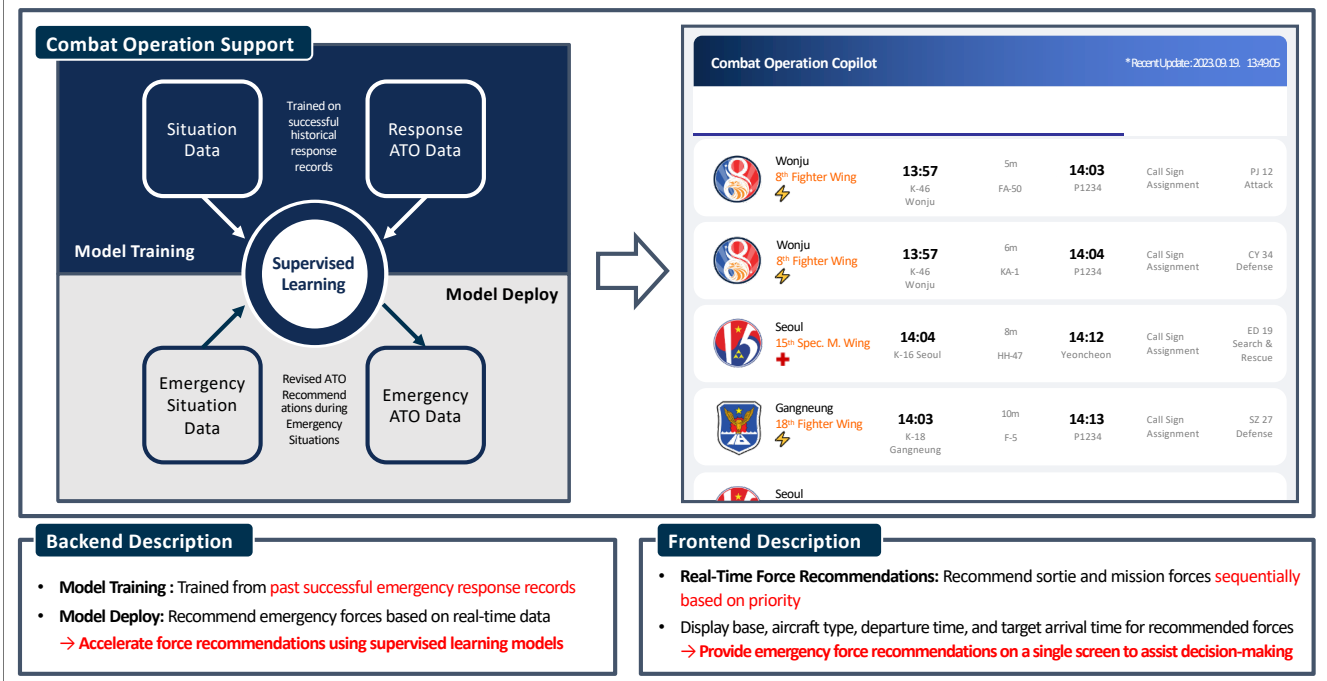
To briefly explain the AI technology applied in this process:

On the left is a **hierarchical deep Q-network (HDQN) model**, a form of reinforcement learning based AI that maximizes tactical effectiveness.

Each layer, represented as a square, corresponds to a single force element -specifically, an aircraft- comprising one mission. These layers learn interactively with one another, improving the success rate of the mission.

On the right, a Monte Carlo Tree Search (MCTS) algorithm employing the UCT (Upper Confidence bound applied to Trees) method performs simulation-based exploration. This approach guarantees sortie generation and bolsters responsiveness to unexpected situations.

2. Implementation of the Idea (5/5) - Combat Operation Support

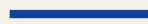


This is the combat operations support function.

By using an **supervised learning based AI model** trained on past response records -w hich incorporate numerous relevant factors- **the system can swiftly generate an emergency ATO** when a crisis occurs.

It presents recommended force information such as the base, aircraft type, scheduled take-off time, and time of arrival at the target, thereby supporting the commander's d ecision-making process.

3



Refinement Plan

Lastly, I would like to discuss our plans for further development.

3. Refinement Plan (1/3) - Trends in AI Military Systems

1. DARPA's [Deep Green] Program (U.S.)

- A system designed by DARPA to support commanders in decision-making
- Features include: ① **Sketch-To-Plan** and **Sketch-To-Decide**: Assists in planning and situational analysis
② **Blitzkrieg**: Conducts simulations ③ **Crystal-Ball**: Analyzes results and identifies key decisions
- Additional tools like **COMPASS** and **AIDA** facilitate visualization of collected data and interaction with commanders
- In 2015, the **Commanders Virtual Staff (CVS)** program was introduced to support decision-making for Army commanders and staff

2. Military Cloud and Database Integration

- Park Jun-kyu et al. (2019) proposed integrating cloud-based solutions and fault-tolerant systems into C4I to enhance interoperability and availability

3. Applicability of AI in ATO Production

- Choi Geun-ha (2020): "AI enhances planning and execution stages by processing collected data and aiding decision-making"
- Clark et al. (2020): AI-based decision-support systems used in war games demonstrated a 3x increase in operational speed and a 100x improvement in task density compared to conventional forces

First, I will outline the additional elements we plan to incorporate, based on relevant overseas and domestic trends.

The U.S. Department of Defense has recognized the effectiveness and necessity of AI-enabled decision support systems and is currently running related programs through DARPA (the Defense Advanced Research Projects Agency). Under ROK-U.S. combined operations, **we will continue to monitor these developments and reflect them in our system's development process** to reinforce interoperability among allied forces.

Moreover, the feasibility of adopting cloud technology in national defense command and control systems has been demonstrated, so once the Air Force's command and control system is integrated with a cloud infrastructure, **we will consider how to operate our system on that cloud platform**.

Lastly, research indicates that AI has high potential in both the planning and execution phases of operations, and that an AI-based command decision system can shorten the operational cycle. By introducing this system, **we aim to validate these capabilities and incorporate them into our operational planning and execution procedures**.

3. Refinement Plan (2/3) - Applicability and Expected Outcomes

1. Shifting Paradigms in Warfare

- The advancement of cutting-edge technologies has expanded the battlefield and shifted the paradigm of warfare from information-based to intelligence-based operations * The AI-powered automated target allocation system, 'GIS Arta,' was deployed during the Russia-Ukraine war
- Beyond enhancing **practical applicability** through data visualization and ease of use, **AI-driven optimal ATO proposals** significantly **contribute to defense innovation**

2. Opportunities to Overcome Defense Challenges

- In modern warfare, survival and victory depend on how quickly battlefield situations are perceived, analyzed, and addressed
- Establishing an AI-based ATO production support system enables a transition to an advanced technological military force, allowing for **effective control over expanded battlefield domains and intelligent warfare methods** that exceed human capabilities

3. Realizing a Decision Support System Optimized for R.O.K. 「Defense Innovation 4.0」

- By combining human command with machine control, dispersed forces can be rapidly reorganized to provide friendly forces with more strategic options, **achieving Decision-Centric Warfare**
- An AI-based ATO production and decision-support system **aligned with the 「Defense Innovation 4.0」 framework** enables the Air Force to quickly achieve optimized operational success in future battlefield environments

Next, let's look at the system's military suitability and anticipated benefits.

With the advancement of cutting-edge technologies, the battlefield is expanding, and the paradigm of warfare is shifting from information warfare to **intelligent warfare**. In the ongoing Russia-Ukraine war, Ukraine has demonstrated significant success by deploying an AI-based automated target distribution system for artillery called GIS Arta.

Not only do **data visualization and user convenience** make this system practical in real-world operations, but the **AI-driven optimal ATO recommendations** will also contribute to defense innovation. By establishing an AI-based support system, we can **quickly comprehend and make decisions** in an expanded battlespace, **enabling us to respond effectively to increasingly intelligent modes of warfare**.

Aligned with "Defense Innovation 4.0" - a core innovation policy under the Republic of Korea's Ministry of National Defense that **seeks to combine human command with machine-driven control for decision-centric warfare**- this approach will help the Air Force **achieve optimized mission success**.

3. Refinement Plan (2-1/3) - Applicability and Expected Outcomes

1. ATO Revision During Combat Operations

- In emergencies like unidentified track detection, real-time team responses and swift commander decisions are essential for timely ATO updates

2. Surpassing the Enemy's OODA-Loop

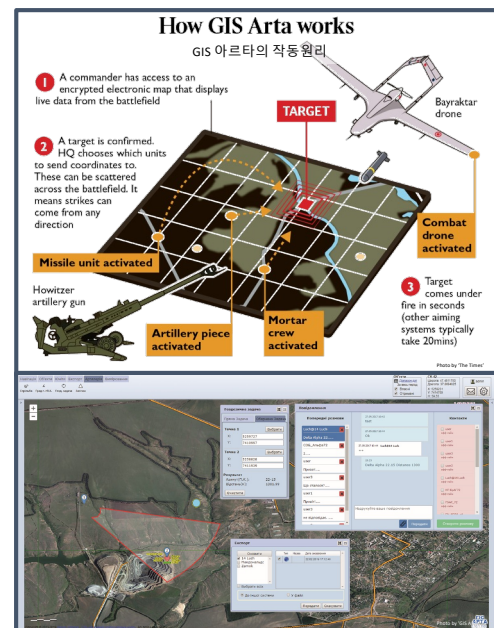
- By maneuvering faster than the enemy within their decision-making cycle (Observe-Orient-Decide-Act), confusion and dilemmas are created within the enemy, ultimately leading to victory
- AI assistance enhances this agility, enabling the Korean Air and Space Operations Center (KAOC) to achieve superior command and control

3. Real-world Application - GIS Arta

- GIS Arta is an advanced Ukrainian military program that integrates data from reconnaissance drones and GPS to confirm enemy positions and uses AI to select the best available attack resources

* Used in annihilating a Russian tactical battalion attempting to cross a river ('22. 05.)

→ The time between enemy position reporting and attack initiation was reduced from 20 minutes to just 1 - 2 minutes



As we near the conclusion of our presentation, let's focus on three key points from this slide that illustrate how an AI-based ATO system can enhance operational effectiveness.

1. ATO Revision During Combat Operations

In high-pressure scenarios (e.g., unexpected aerial threats), real-time collaboration and rapid decision-making are vital. Our system **enables swift ATO modifications, ensuring agile and effective responses.**

2. Surpassing the Enemy's OODA Loop

By acting faster than the opponent in their Observe–Orient–Decide–Act cycle, we **create confusion and maintain a decisive edge.** AI support further enhances our ability to make quick, informed decisions.

3. Real-World Application: GIS Arta

Ukraine's GIS Arta showcases how AI can drastically **reduce the time between identifying enemy positions and initiating attacks**—from around 20 minutes to just 1–2 minutes—highlighting the tangible benefits of AI-powered mission support.

These examples underscore the potential for leveraging AI to optimize mission planning and execution.

Next, we'll share *insights from field operators regarding their perspectives* on this concept in our "Field Interview" segment.

3. Refinement Plan (3/3) - Field Interviews (MAAP Planning Team)

“If a program can integrate ATO mission planning with real-time force flight control, it will surely help enable more effective and agile KAO operations!”

AFOC CPD Maj Jeong and the MAAP Team
Operations Plans Division

3. Refinement Plan (3-1/3) - Field Interviews (ATO Production Team)

“Ultimately, faster ATO production will reduce unnecessary mission plan re-verifications by personnel and minimize the need for post-dissemination modifications by the operations team, making this approach very beneficial.”

AFOC CPD Capt Choi
ATO Production Team
(Hackathon Mentor)

“If this idea is implemented, the time required for target and resource allocation can be drastically reduced. Furthermore, faster wartime planning would significantly help in independent planning or post-operational control transitions.”

AFOC CPD Grade 6 Civ Kim
ATO Production Team

3. Refinement Plan (3-2/3) - Field Interviews (Combat Operations Team)

“**The real-time force recommendation system idea is excellent.** A program that can adapt to various complex scenarios is necessary. Until now, suitable aircraft for missions have been manually selected. An automated program like this would greatly benefit combat operations.”

AFOC COD Lt. Col Park
Combat Operations Team

“**This idea is fantastic!** From a commander’s perspective, the UI displaying only the necessary information is very impressive.”

AFOC COD Lt. Col Han
Combat Operations Team

“When revising the ATO, there are many considerations when deploying new forces or changing missions. The ability to recommend multiple force options makes this even more useful!”

AFOC COD MSgt Park
Combat Operations Team

4

Q & A

“What an incredible system... Let’s do this.”

AFOC A6 Lt. Col. Hong
Systems Division, Command and Communications Department

Thank You

RunTime Error!
R.O.K Air Force Hackathon

Reference

1. **Choi, Geun-ha.** (2020). *Conceptual Design of AI-based Tactical Echelon Decision Support System Using Functional Analysis Method*.
2. **Lee, Chang-eun, et al.** (2021). *Trends in AI-based Military Staff Technology for Supporting Commanders' Decision-Making*. Electronics and Telecommunications Research Institute, Korea.
3. **Defense Science Research Institute.** (2021). *Trends and Development Plans for Intelligent Decision Support Technologies Utilizing AI and Digital Twins in the Battlefield*. Defense Science and Technology+, Vol. 252, 2nd Quarter, Korea.
4. **Defense Technology Promotion Research Institute.** (2022). *Forecasting Future Defense Technologies Using Big Data*.
5. **Jung, Young-joo.** (2022). *Development of AI-based ATO Planning Systems: Focused on Automating Combat Planning*. National Defense Policy Research.
6. **Yang, Wook.** (2022). *Future Warfare of Decision-Centric Operations through Mosaic Warfare*. ASAN REPORT.
7. **Kim, Jong-bae & Park, Jin-ho.** (2019). *Strategies and Development Directions for Military Cloud Integration*. Journal of IT Policy and Management, Korea.
8. **Park, Jun-kyu, et al.** (2019). *Possibility of Cloud Adoption in Defense Command and Control Systems through Fault Tolerance and Interoperability*. Korea.
9. **Song, Ji-yoo.** (2022). 'Uber Technology' Played a Crucial Role in Russia's 'Death Bridge' Incident, Resulting in 1500 Russian Troops' Demise. Money Today, May 16. <https://news.mt.co.kr/mtview.php?no=2022051612250249841>